

EA-0012 - AQELYN Engineering Archive

AQELYN - EA-0012 Engineering Archive

IS-012 - AQELYN Asset & Configuration Governance Engine

****Archive ID:**** EA-0012

****Implementation Specification:**** IS-012

****Component:**** AQELYN Asset & Configuration Governance Engine

****Project:**** AQELYN

****System Type:**** Cyber Security Operating Environment

****Status:**** COMPLETE

****Repository Impact:**** No top-level repository structure changes

****Breaking Changes:**** None

****Engineering Phase:**** Phase 3

****Predecessor Archives:**** EA-0001 through EA-0011

****Next Specification:**** IS-013

Document Control

| Field | Value |

---|---

| Document | Engineering Archive EA-0012 |

| Specification | IS-012 - AQELYN Asset & Configuration Governance Engine |

| Publication Format | Markdown, PDF, HTML, ZIP |

| Source of Truth | MD/EA-0012.md |

| Archive Rule | Implementation Specification -> Engineering Archive -> Continue |

| Repository Rule | Fixed repository structure; no redesign |

| Completion State | IS-012 complete; EA-0012 generated |

1. Engineering Context

AQELYN is being built as a modular Cyber Security Operating Environment.

Completed engineering archive chain:

| Engineering Archive | Implementation Specification | Component |

---|---|---

| EA-0001 | IS-001 | AQELYN Kernel |

| EA-0002 | IS-002 | Universal Object Model |

| EA-0003 | IS-003 | AQELYN Event Bus |

EA-0004	IS-004	AQELYN Evidence Engine
EA-0005	IS-005	AQELYN Knowledge Graph
EA-0006	IS-006	AQELYN Trust Engine
EA-0007	IS-007	AQELYN Mission Engine
EA-0008	IS-008	AQELYN Workflow Engine
EA-0009	IS-009	AQELYN Policy Engine
EA-0010	IS-010	AQELYN Compliance & Governance Engine
EA-0011	IS-011	AQELYN Identity & Access Governance Engine
EA-0012	IS-012	AQELYN Asset & Configuration Governance Engine

The fixed repository structure remains:

```
AQELYN/  
■■■■ archive/  
■■■■ blueprint/  
■■■■ docs/  
■■■■ src/  
■■■■ tests/  
■■■■ tools/  
■■■■ build/  
■■■■ releases/  
■■■■ scripts/  
■■■■ assets/  
■■■■ examples/  
■■■■ plugins/  
■■■■ sdk/  
■■■■ api/  
■■■■ README.md
```

The engineering rule remains:

```
Finish Implementation Specification  
↓  
Generate Engineering Archive  
↓  
Continue
```

No Engineering Archive may be skipped.

2. IS-012 Specification Identity

```
Specification ID: IS-012  
Name: AQELYN Asset & Configuration Governance Engine  
Engineering Archive Target: EA-0012  
Project: AQELYN  
System Type: Cyber Security Operating Environment  
Status: Complete  
Predecessor: IS-011 - AQELYN Identity & Access Governance Engine
```

3. Purpose

The AQELYN Asset & Configuration Governance Engine manages cyber assets, system inventory, configuration state, ownership, baseline compliance, drift detection, exposure context, and asset governance across AQELYN.

It answers:

```
What assets exist?  
Who owns them?  
Where are they?  
What configuration do they run?  
Are they compliant with baseline?  
Did configuration drift occur?  
Which assets are exposed?
```

Which assets support missions?
Which evidence proves their state?

4. Mission

The engine shall provide:

Asset inventory
Configuration inventory
Ownership mapping
Baseline governance
Configuration drift detection
Asset classification
Exposure tracking
Asset-to-mission mapping
Asset-to-policy mapping
Asset-to-evidence mapping
Asset risk context
Integration with compliance, identity, trust, policy, workflow, evidence, and knowledge graph engines

5. Scope

5.1 In Scope

Asset registry
Configuration registry
Asset ownership
Asset classification
Asset lifecycle
Configuration baseline management
Configuration drift detection
Asset exposure state
Asset criticality scoring
Asset-to-identity mapping
Asset-to-policy mapping
Asset-to-evidence mapping
Asset-to-mission mapping
Asset governance events
Asset review workflows

5.2 Out of Scope

Full CMDB replacement
Endpoint detection and response implementation
Vulnerability scanner implementation
Cloud provider replacement
Configuration management tool replacement
Patch deployment engine
Physical inventory logistics system

6. Dependencies

IS-012 depends on:

IS-001 AQELYN Kernel
IS-002 Universal Object Model
IS-003 AQELYN Event Bus
IS-004 AQELYN Evidence Engine
IS-005 AQELYN Knowledge Graph
IS-006 AQELYN Trust Engine
IS-007 AQELYN Mission Engine
IS-008 AQELYN Workflow Engine
IS-009 AQELYN Policy Engine
IS-010 AQELYN Compliance & Governance Engine

IS-011 AQELYN Identity & Access Governance Engine

7. High-Level Architecture

AQELYN Asset & Configuration Governance Engine

-
- Asset Registry
- Configuration Registry
- Asset Discovery Normalizer
- Ownership & Stewardship Manager
- Asset Classification Service
- Baseline Manager
- Configuration Drift Detector
- Exposure Context Service
- Asset Criticality Engine
- Asset Evidence Binder
- Asset Review Workflow Connector
- Asset Reporting Service
- Asset Event Publisher

8. Functional Requirements

FR-012-001 - Asset Registry

The system shall maintain governed records for:

- Servers
- Endpoints
- Network devices
- Cloud resources
- Containers
- Kubernetes workloads
- Applications
- Databases
- APIs
- Identity systems
- Security tools
- SaaS services
- Data stores
- IoT devices
- OT devices

FR-012-002 - Configuration Registry

The system shall maintain configuration state for governed assets.

Each configuration record shall include:

- configuration_id
- asset_id
- source
- configuration_type
- observed_value
- expected_value
- baseline_id
- observed_at
- confidence
- evidence_links
- status

FR-012-003 - Asset Ownership

Each asset shall have an accountable owner or governance exception.

Ownership metadata shall include:

```
owner_id
steward_id
business_unit
technical_owner
security_owner
mission_owner
review_frequency
```

FR-012-004 - Asset Classification

The system shall classify assets by:

```
asset_type
environment
sensitivity
criticality
exposure
mission_relevance
data_classification
regulatory_scope
trust_zone
```

FR-012-005 - Baseline Management

The system shall define and maintain configuration baselines.

Baselines shall include:

```
baseline_id
name
scope
required_controls
expected_configurations
policy_links
compliance_links
effective_date
version
status
```

FR-012-006 - Drift Detection

The system shall detect configuration drift between observed and expected state.

Drift states:

```
no_drift
minor_drift
major_drift
critical_drift
unknown
accepted_exception
```

FR-012-007 - Exposure Tracking

The system shall track asset exposure context.

Examples:

```
internet_exposed
internal_only
partner_accessible
privileged_accessible
mission_exposed
public_api
unclassified
```

FR-012-008 - Asset Criticality

The system shall calculate asset criticality using:

```
mission dependency
business function
data sensitivity
exposure state
identity access level
compliance scope
trust score
known findings
```

FR-012-009 - Evidence Binding

Asset and configuration states shall reference immutable evidence objects from the AQELYN Evidence Engine.

FR-012-010 - Governance Review

The system shall support asset review workflows for:

```
ownership validation
classification review
baseline compliance review
configuration exception review
asset retirement review
exposure review
```

9. Non-Functional Requirements

The engine shall satisfy:

```
Traceability
Auditability
Event-driven synchronization
Configuration integrity
Evidence-backed state
Explainable drift decisions
Scalable asset graph processing
Tenant-aware governance
Backward compatibility
Repository stability
```

10. Core Governance Flow

```
Asset discovered
↓
Asset normalized
↓
Owner assigned
↓
Classification applied
↓
Configuration observed
↓
Baseline evaluated
↓
Drift detected
↓
Risk and criticality calculated
↓
Evidence linked
↓
Governance workflow triggered
↓
Asset event published
```

11. Internal Component Architecture

The Asset & Configuration Governance Engine consists of modular services communicating through the AQELYN Event Bus.

AQELYN Asset & Configuration Governance Engine

-
- Asset Registry
- Configuration Registry
- Discovery Normalization Service
- Ownership & Stewardship Manager
- Asset Classification Engine
- Baseline Manager
- Configuration Drift Detector
- Exposure Context Service
- Asset Criticality Engine
- Asset Evidence Service
- Governance Connector
- Reporting Service
- Event Publisher

Each component shall be independently deployable, versioned, and testable.

12. Component Specifications

12.1 Asset Registry

The Asset Registry maintains the authoritative inventory of governed assets.

Supported asset categories:

- Physical Servers
- Virtual Machines
- Cloud Instances
- Containers
- Kubernetes Nodes
- Applications
- Databases
- Network Devices
- Firewalls
- Load Balancers
- Storage Systems
- Identity Systems
- Security Appliances
- SaaS Platforms
- IoT Devices
- OT Systems

Responsibilities:

- Asset registration
- Lifecycle tracking
- Ownership association
- Metadata management
- Relationship management

12.2 Configuration Registry

Stores observed and expected configuration state.

Supported configuration domains:

- Operating Systems
- Network Configuration
- Cloud Configuration
- Container Configuration
- Application Configuration
- Database Configuration
- Security Configuration
- Identity Configuration
- Policy Configuration

Each configuration item shall be versioned and timestamped.

12.3 Discovery Normalization Service

Normalizes inventory from heterogeneous discovery sources.

Supported discovery sources include:

- Cloud APIs
- CMDB exports
- Network discovery
- Container platforms
- Endpoint agents
- Identity systems
- Infrastructure-as-Code repositories
- Manual registration

Normalization ensures a consistent Universal Object Model representation.

12.4 Ownership & Stewardship Manager

Maintains governance ownership.

Supported ownership roles:

- Business Owner
- Technical Owner
- Security Owner
- Mission Owner
- Data Steward
- Application Owner
- Infrastructure Owner

Supports delegated ownership and governance exceptions.

12.5 Asset Classification Engine

Assigns classifications using configurable policy rules.

Classification dimensions:

- Criticality
- Confidentiality
- Integrity
- Availability
- Environment
- Mission Importance
- Exposure
- Regulatory Scope
- Trust Zone

12.6 Baseline Manager

Maintains approved configuration baselines.

Capabilities:

- Baseline creation
- Baseline versioning
- Baseline comparison
- Baseline retirement
- Baseline approval
- Policy linkage

12.7 Configuration Drift Detector

Continuously compares observed configuration against approved baselines.

Drift severity:

- None
- Informational
- Minor
- Major
- Critical

Drift events may trigger workflow automation.

12.8 Exposure Context Service

Maintains current exposure posture.

Exposure examples:

- Internet Facing
- Internal
- Partner Connected
- Air-Gapped
- Restricted
- Mission Critical
- Development
- Production

12.9 Asset Criticality Engine

Calculates asset criticality based on:

- Mission dependency
- Business importance
- Exposure
- Trust score
- Compliance impact
- Identity privilege level
- Known findings
- Configuration health

Output levels:

- Low
- Moderate
- High
- Critical

12.10 Asset Evidence Service

Associates asset state with immutable evidence.

Evidence sources:

- Discovery records
- Configuration snapshots
- Baseline approvals

```
Ownership approvals
Policy evaluations
Workflow history
Compliance reviews
```

13. Universal Object Model Extensions

The following domain objects extend IS-002.

13.1 Asset

```
Asset:
asset_id
asset_type
owner
classification
lifecycle_state
trust_score
criticality
```

13.2 ConfigurationItem

```
ConfigurationItem:
configuration_id
asset_id
source
baseline
observed_state
expected_state
drift_status
```

13.3 Baseline

```
Baseline:
baseline_id
version
scope
effective_date
status
```

13.4 AssetRelationship

```
AssetRelationship:
relationship_id
source_asset
target_asset
relationship_type
```

13.5 DriftFinding

```
DriftFinding:
finding_id
asset_id
severity
baseline
evidence
```

13.6 AssetReview

```
AssetReview:
review_id
reviewer
decision
timestamp
```

14. Knowledge Graph Integration

Relationships include:

```
Asset
↓
owned_by
↓
Identity
↓
managed_by
↓
Organization

Asset
↓
implements
↓
Mission

Asset
↓
protected_by
↓
Policy

Asset
↓
supported_by
↓
Evidence
```

15. Event Bus Integration

The engine publishes standardized events.

15.1 Asset Events

```
asset.discovered
asset.registered
asset.updated
asset.retired
asset.deleted
```

15.2 Configuration Events

```
configuration.observed
configuration.changed
configuration.baseline.updated
configuration.drift.detected
```

15.3 Governance Events

```
asset.review.completed
asset.classification.changed
asset.owner.changed
asset.exception.approved
```

15.4 Risk Events

```
asset.criticality.changed
asset.exposure.changed
asset.risk.updated
```

16. Evidence Engine Integration

Evidence references include:

- Discovery Evidence
- Configuration Snapshots
- Baseline Approvals
- Ownership Records
- Review Records
- Workflow Decisions

Only immutable evidence identifiers shall be referenced.

17. Policy Engine Integration

The Policy Engine determines:

- Required asset classifications
- Required baselines
- Required owners
- Required reviews
- Configuration exceptions
- Review frequency

18. Compliance & Governance Integration

IS-010 consumes:

- Baseline compliance
- Configuration drift
- Ownership completeness
- Exposure posture
- Asset inventory
- Evidence completeness

19. Identity Governance Integration

IS-011 supplies:

- Asset owners
- Technical owners
- Security owners
- Mission owners
- Privileged identities

The Asset Governance Engine never duplicates identity records; it references governed identities maintained by IS-011.

20. Workflow Engine Integration

Example workflow:

- Asset Discovered
- ↓
- Normalization
- ↓
- Ownership Assignment
- ↓
- Classification
- ↓
- Baseline Evaluation

```
↓  
Drift Analysis  
↓  
Risk Calculation  
↓  
Evidence Recording  
↓  
Governance Review
```

21. Trust Engine Integration

Asset confidence is calculated using:

```
Discovery Confidence  
Evidence Confidence  
Configuration Confidence  
Identity Confidence  
Policy Confidence
```

Produces:

```
Asset Trust Score
```

22. Mission Engine Integration

Mission policies may define:

```
Required Assets  
Required Baselines  
Required Configuration State  
Required Availability  
Required Criticality
```

Mission readiness depends on governed asset state.

23. Public APIs

23.1 Asset API

```
GET /assets  
POST /assets  
GET /assets/{id}  
PUT /assets/{id}
```

23.2 Configuration API

```
GET /configurations  
POST /configurations  
GET /configurations/{id}  
PUT /configurations/{id}
```

23.3 Baseline API

```
GET /baselines  
POST /baselines  
PUT /baselines/{id}
```

23.4 Drift API

```
GET /drift
GET /drift/{asset}
POST /drift/recalculate
```

23.5 Governance API

```
GET /asset/reviews
POST /asset/reviews
GET /asset/governance
```

24. Repository Impact

No repository redesign.

Expected implementation:

```
AQELYN/
■■■ src/
■ ■■■ asset_governance/
■■■ tests/
■ ■■■ asset_governance/
■■■ docs/
■ ■■■ asset_governance/
■■■ api/
■ ■■■ asset_governance/
■■■ archive/
```

The permanent AQELYN repository structure remains unchanged.

25. Security Architecture

The AQELYN Asset & Configuration Governance Engine is a Tier-1 governance subsystem responsible for maintaining authoritative knowledge of enterprise assets and their configuration state.

Every asset decision shall be:

```
Policy-driven
Evidence-backed
Fully auditable
Version controlled
Traceable
Explainable
```

Configuration governance shall never rely on undocumented state.

25.1 Security Principles

The engine shall implement:

```
Zero Trust
Configuration Integrity
Least Privilege
Defense in Depth
Immutable Evidence
Continuous Validation
Asset Accountability
Separation of Duties
Security by Design
```

25.2 Asset Authorization Model

Only authorized governance roles may modify asset records.

Supported governance roles:

Role Responsibility
--- ---
Asset Administrator Asset lifecycle management
Technical Owner Configuration ownership
Security Owner Security configuration approval
Business Owner Business accountability
Mission Owner Mission dependency approval
Compliance Officer Compliance validation
Automation Service Approved automated synchronization

Authorization decisions shall be enforced through the AQELYN Policy Engine.

25.3 Configuration Integrity

Configuration records shall maintain:

Unique Identifier
Version
Baseline Reference
Evidence References
Observation Timestamp
Source System
Integrity Status

Historical configuration records shall remain immutable.

25.4 Baseline Protection

Approved baselines shall support:

Version control
Approval history
Policy linkage
Evidence linkage
Retirement history
Exception tracking

Only approved baselines may be used for compliance evaluation.

26. Asset Lifecycle

Every governed asset follows a controlled lifecycle.

Discovered
↓
Registered
↓
Classified
↓
Assigned Owner
↓
Operational
↓
Reviewed
↓
Maintained
↓
Retired

↓
Archived

26.1 Configuration Lifecycle

Observed
↓
Normalized
↓
Baseline Matched
↓
Validated
↓
Monitored
↓
Drift Detected
↓
Remediated
↓
Archived

26.2 Baseline Lifecycle

Draft
↓
Reviewed
↓
Approved
↓
Published
↓
Applied
↓
Revised
↓
Retired

27. Risk Evaluation

Asset governance shall evaluate:

Asset Criticality
Configuration Drift
Exposure Level
Mission Dependency
Identity Privilege
Compliance Scope
Trust Score
Known Findings
Evidence Quality

Risk classifications:

Low
Moderate
High
Critical

Critical assets shall require enhanced governance workflows.

28. Continuous Governance

The engine shall continuously evaluate:

Asset inventory completeness
Ownership validity

- Configuration drift
- Baseline compliance
- Exposure changes
- Mission dependencies
- Unauthorized configuration changes
- Missing evidence
- Retired assets still in service
- Unmanaged assets

Detected issues shall generate governance events and remediation workflows where configured.

29. Audit & Reporting

29.1 Executive Reports

- Enterprise Asset Inventory
- Critical Asset Summary
- Configuration Compliance
- Exposure Overview

29.2 Operational Reports

- Configuration Drift
- Missing Owners
- Retired Assets
- Discovery Statistics

29.3 Compliance Reports

- Baseline Compliance
- Configuration Exceptions
- Asset Coverage
- Evidence Completeness

29.4 Engineering Reports

- Asset Relationships
- Configuration History
- Baseline Evolution
- Governance Workflow Metrics

30. Failure Handling

30.1 Missing Owner

Status:
Governance Incomplete

Action:
Escalate for ownership assignment

30.2 Missing Baseline

Status:
Configuration Unverified

Action:
Create governance review task

30.3 Discovery Failure

Status:
Observation Failed

Action:
Retry according to discovery policy

30.4 Missing Evidence

Status:
Evidence Incomplete

Action:
Manual governance review required

30.5 Event Bus Interruption

Asset governance events shall be queued until successful delivery is restored.

31. Performance Requirements

The engine shall support:

Continuous asset synchronization
High-volume configuration ingestion
Parallel drift analysis
Incremental baseline evaluation
Asynchronous governance workflows
Large-scale reporting

Asset discovery shall not interrupt governance processing.

32. Scalability Requirements

The architecture shall support:

Millions of assets
Millions of configuration records
Large enterprise inventories
Multi-cloud deployments
Hybrid infrastructure
Multi-tenant governance
Distributed processing

33. Testing Strategy

33.1 Unit Testing

Validate:

Asset registry
Configuration registry
Classification engine
Drift detector
Criticality engine
Baseline manager
Evidence binding

33.2 Integration Testing

Verify interaction with:

- AQELYN Kernel
- Event Bus
- Evidence Engine
- Knowledge Graph
- Trust Engine
- Workflow Engine
- Policy Engine
- Compliance Engine
- Identity Governance Engine
- Mission Engine

33.3 System Testing

Validate complete scenarios including:

- Asset discovery
- Asset normalization
- Ownership assignment
- Configuration observation
- Baseline evaluation
- Drift detection
- Governance review
- Compliance reporting

33.4 Security Testing

Verify:

- Authorization
- Configuration integrity
- Baseline protection
- Audit trail completeness
- Evidence integrity
- Governance workflow security

33.5 Regression Testing

Ensure IS-001 through IS-011 continue to operate without behavioral changes introduced by IS-012.

34. Acceptance Criteria

IS-012 shall be considered complete when:

- Asset registry is defined.
- Configuration registry is defined.
- Ownership model is documented.
- Asset classification is defined.
- Baseline management is documented.
- Drift detection model is defined.
- Exposure model is documented.
- Risk evaluation model is documented.
- Integration with IS-001 through IS-011 is defined.
- Repository structure remains unchanged.
- Testing strategy is documented.

35. Repository Validation

This specification introduces no changes to the approved repository structure.

Expected implementation:

```
AQELYN/  
■■■ src/asset_governance/  
■■■ tests/asset_governance/  
■■■ docs/asset_governance/  
■■■ api/asset_governance/  
■■■ archive/
```

The AQELYN repository remains stable and unchanged.

36. Completion Summary

IS-012 - AQELYN Asset & Configuration Governance Engine establishes a comprehensive governance capability for enterprise assets and configuration state by:

```
Maintaining authoritative asset and configuration inventories.  
Governing ownership, classification, and criticality.  
Managing approved configuration baselines.  
Detecting and evaluating configuration drift.  
Tracking exposure and mission relevance.  
Integrating with identity, policy, trust, workflow, evidence, compliance, and mission engines.  
Preserving repository stability and backward compatibility.
```

37. Specification Status

```
Specification ID : IS-012  
Title : AQELYN Asset & Configuration Governance Engine  
Status : COMPLETE  
Engineering Archive : READY FOR GENERATION  
Next Artifact : EA-0012
```

Engineering workflow status:

```
EA-0001 COMPLETE  
EA-0002 COMPLETE  
EA-0003 COMPLETE  
EA-0004 COMPLETE  
EA-0005 COMPLETE  
EA-0006 COMPLETE  
EA-0007 COMPLETE  
EA-0008 COMPLETE  
EA-0009 COMPLETE  
EA-0010 COMPLETE  
EA-0011 COMPLETE  
IS-012 COMPLETE  
EA-0012 READY FOR GENERATION
```

38. EA-0012 Engineering Objective

The objective of IS-012 was to introduce a dedicated Asset & Configuration Governance Engine that enables AQELYN to govern asset inventory, configuration state, ownership, classification, baselines, drift, exposure, criticality, and evidence-backed configuration accountability.

The engine extends AQELYN governance from identity and access into the governed state of cyber assets and the configurations that determine their security posture.

39. EA-0012 Engineering Summary

The implementation specification defines a modular subsystem responsible for:

```
Asset registry management
Configuration registry management
Discovery normalization
Ownership and stewardship management
Asset classification
Baseline management
Configuration drift detection
Exposure context tracking
Asset criticality calculation
Evidence binding for asset and configuration state
Asset review workflows
Asset governance reporting
```

The engine integrates with all previously completed AQELYN engines while preserving architectural modularity.

40. Major Engineering Decisions

40.1 Decision 1 - Dedicated Asset Governance Engine

Asset and configuration governance responsibilities are implemented as a standalone engine rather than embedded in Compliance, Policy, or Workflow.

Rationale:

```
Clear separation of asset state from policy evaluation.
Independent lifecycle and scaling.
Better support for discovery normalization and baseline management.
Improved traceability of ownership, exposure, and configuration drift.
```

40.2 Decision 2 - Asset-to-Configuration Separation

The engine explicitly separates asset records from configuration records.

Benefits:

```
One asset can have many configuration observations.
Configuration history can remain immutable.
Baseline comparisons remain explainable.
Drift analysis does not overwrite asset identity.
```

40.3 Decision 3 - Evidence-Backed Configuration State

Configuration and asset state reference immutable evidence maintained by the AQELYN Evidence Engine.

Benefits:

```
Configuration state becomes auditable.
Baseline compliance is evidence-backed.
Asset reviews can be traced.
Compliance requirements can consume asset evidence.
```

40.4 Decision 4 - Event-Driven Asset Governance

Asset, configuration, drift, governance, and risk state changes are published through the AQELYN Event Bus.

Examples include:

```
asset.discovered
asset.registered
asset.updated
configuration.observed
configuration.drift.detected
asset.review.completed
```

```
asset.criticality.changed
asset.exposure.changed
asset.risk.updated
```

This maintains loose coupling between AQELYN engines.

40.5 Decision 5 - Universal Object Model Extension

New domain objects introduced include:

```
Asset
ConfigurationItem
Baseline
AssetRelationship
DriftFinding
AssetReview
```

These extend the Universal Object Model without modifying existing object definitions.

41. Architectural Integration Summary

| Engine | Integration |

|---|---|

| IS-001 Kernel | Runtime lifecycle and service registration |

| IS-002 Universal Object Model | Asset and configuration objects |

| IS-003 Event Bus | Asset, configuration, drift, governance, and risk events |

| IS-004 Evidence Engine | Discovery, configuration, baseline, ownership, and review evidence |

| IS-005 Knowledge Graph | Asset, identity, policy, evidence, mission, and relationship graph |

| IS-006 Trust Engine | Asset trust, evidence confidence, configuration confidence |

| IS-007 Mission Engine | Mission-required assets, baselines, state, and availability |

| IS-008 Workflow Engine | Ownership assignment, drift remediation, review workflows |

| IS-009 Policy Engine | Classification, baseline, owner, review, and exception rules |

| IS-010 Compliance Engine | Baseline compliance, drift, exposure, asset inventory |

| IS-011 Identity Governance Engine | Owners, stewards, privileged identities, mission owners |

No existing engine required redesign.

42. Repository Impact Summary

Repository structure remains unchanged.

Implementation is expected within existing project directories, including:

```
AQELYN/
■■■ src/asset_governance/
■■■ tests/asset_governance/
■■■ api/asset_governance/
■■■ docs/asset_governance/
■■■ archive/
```

No top-level directories were added, removed, or renamed.

43. Security Impact Summary

The specification introduces asset-governance-specific security controls:

- Policy-driven asset authorization
- Configuration integrity
- Baseline protection
- Asset accountability
- Evidence-backed configuration state
- Immutable historical configuration records
- Continuous validation
- Separation of duties for asset and baseline changes

No reduction in the security posture of existing components was identified.

44. Capabilities Added

The engine enables AQELYN to support:

- Asset inventory
- Configuration inventory
- Discovery normalization
- Ownership governance
- Stewardship governance
- Asset classification
- Configuration baselines
- Drift detection
- Exposure tracking
- Criticality scoring
- Asset-to-identity mapping
- Asset-to-policy mapping
- Asset-to-evidence mapping
- Asset-to-mission mapping
- Governance reviews
- Compliance reporting
- Mission readiness context

45. Risks Identified

Risk Mitigation	
--- ---	
Incomplete asset discovery	Multi-source discovery normalization and confidence scoring
Duplicate asset records	Normalization and Knowledge Graph relationship management
Missing ownership	Governance review workflows and escalation
Baseline misconfiguration	Versioning, approval history, and policy linkage
Drift false positives	Severity classification and exception handling
High-volume configuration ingestion	Parallel drift analysis and incremental evaluation
Evidence gaps	Immutable evidence references and manual review workflow
Exposure misclassification	Policy-driven exposure review and evidence-backed classification

No critical architectural risks were identified that require redesign.

46. Verification Summary

The specification defines verification for:

```
Unit testing
Integration testing
System testing
Security testing
Regression testing
```

Acceptance criteria cover asset registry, configuration registry, ownership, classification, baseline management, drift detection, exposure model, risk evaluation, integration with IS-001 through IS-011, repository validation, and testing documentation.

47. Engineering Principles Confirmed

The implementation complies with established AQELYN principles:

```
Modular architecture
Event-driven communication
Immutable evidence references
Traceability
Explainability
Security by design
Repository stability
Backward compatibility
Governance-before-archive discipline
```

48. Dependencies

Required:

```
EA-0001 through EA-0011
IS-001 through IS-011
```

Enables:

```
IS-013 and subsequent asset-dependent components
```

49. Completion Record

```
Engineering Archive : EA-0012
Implementation Specification : IS-012
Title : AQELYN Asset & Configuration Governance Engine
Engineering Status : COMPLETE
Repository Status : UNCHANGED
Architecture Status : EXTENDED
Backward Compatibility : MAINTAINED
Engineering Rule :
IS Completed
↓
EA Generated
↓
Continue
```

50. Archive Index Update

```
EA-0001 IS-001 AQELYN Kernel
EA-0002 IS-002 Universal Object Model
EA-0003 IS-003 AQELYN Event Bus
EA-0004 IS-004 AQELYN Evidence Engine
EA-0005 IS-005 AQELYN Knowledge Graph
EA-0006 IS-006 AQELYN Trust Engine
EA-0007 IS-007 AQELYN Mission Engine
EA-0008 IS-008 AQELYN Workflow Engine
```



```
EA-0009 IS-009 AQELYN Policy Engine
EA-0010 IS-010 AQELYN Compliance & Governance Engine
EA-0011 IS-011 AQELYN Identity & Access Governance Engine
EA-0012 IS-012 AQELYN Asset & Configuration Governance Engine
```

51. Engineering Phase Status

Completed Engineering Archives : EA-0001 through EA-0012

Current Status:
EA-0012 COMPLETE

Next Implementation Specification:
IS-013

EA-0012 is completed and archived. The engineering workflow is consistent with the project rule:

Implementation Specification -> Engineering Archive -> Continue

From this point onward, the next engineering artifact is IS-013.

52. Engineering Archive Publication Standard

EA-0012 follows the AQELYN Engineering Archive Publication Standard.

Required package structure:

```
EA-xxxx/
■
■ diagrams/
■ ■ ■ ■ Architecture.svg
■ ■ ■ ■ Component.svg
■ ■ ■ ■ Workflow.svg
■ ■ ■ ■ EventFlow.svg
■ ■ ■ ■ Integration.svg
■
■ ■ ■ ■ examples/
■ ■ ■ ■ example_*.md
■
■ ■ ■ ■ HTML/
■ ■ ■ ■ index.html
■ ■ ■ ■ styles.css
■ ■ ■ ■ assets/
■
■ ■ ■ ■ images/
■
■ ■ ■ ■ journal/
■ ■ ■ ■ Engineering_Journal.md
■
■ ■ ■ ■ manifest/
■ ■ ■ ■ manifest.json
■
■ ■ ■ ■ MD/
■ ■ ■ ■ EA-xxxx.md
■
■ ■ ■ ■ PDF/
■ ■ ■ ■ EA-xxxx.pdf
■
■ ■ ■ ■ requirements/
■ ■ ■ ■ Requirements_Matrix.md
■
■ ■ ■ ■ traceability/
■ ■ ■ ■ Traceability_Matrix.md
■
■ ■ ■ ■ README.md
```

The master Markdown is the source of truth. The PDF and HTML are generated from the same master Markdown and must not omit sections.

53. Requirements Matrix

Requirement ID	Requirement	Evidence in Archive	Status
FR-012-001	Maintain governed asset records	Sections 8, 12	Complete
FR-012-002	Maintain configuration state	Sections 8, 12, 13	Complete
FR-012-003	Maintain asset ownership	Sections 8, 12, 25	Complete
FR-012-004	Classify assets	Sections 8, 12	Complete
FR-012-005	Define and maintain baselines	Sections 8, 12, 26	Complete
FR-012-006	Detect configuration drift	Sections 8, 12, 26, 27	Complete
FR-012-007	Track asset exposure	Sections 8, 12, 28	Complete
FR-012-008	Calculate asset criticality	Sections 8, 12, 27	Complete
FR-012-009	Bind asset/configuration state to evidence	Sections 8, 16, 40	Complete
FR-012-010	Support governance reviews	Sections 8, 20, 28	Complete
NFR-012-001	Traceability	Sections 9, 14, 16, 49	Complete
NFR-012-002	Auditability	Sections 9, 15, 29, 33	Complete
NFR-012-003	Event-driven synchronization	Sections 9, 15, 31	Complete
NFR-012-004	Configuration integrity	Sections 9, 25	Complete
NFR-012-005	Evidence-backed state	Sections 9, 16, 40	Complete
NFR-012-006	Repository stability	Sections 24, 35, 42	Complete

54. Traceability Matrix

Source	Target	Relationship
IS-012 Purpose	EA-0012 Objective	Defines why the engine exists
Asset Registry	FR-012-001	Implements governed asset inventory
Configuration Registry	FR-012-002	Implements configuration state tracking
Ownership & Stewardship Manager	FR-012-003	Implements accountable ownership
Asset Classification Engine	FR-012-004	Implements classification model
Baseline Manager	FR-012-005	Implements baseline lifecycle
Drift Detector	FR-012-006	Implements drift detection
Exposure Context Service	FR-012-007	Tracks exposure state
Asset Criticality Engine	FR-012-008	Calculates criticality
Asset Evidence Service	FR-012-009	Binds state to evidence
Governance Connector	FR-012-010	Triggers review workflows
Evidence Engine Integration	Evidence-backed state	References immutable evidence
Policy Engine Integration	Authorization and baseline rules	Determines classifications, baselines, reviews
Compliance Integration	IS-010	Supplies baseline compliance and asset coverage
Identity Governance Integration	IS-011	Supplies asset owners and privileged identities
Event Bus Integration	NFR-012-003	Publishes asset and drift events
Repository Validation	Repository Standard	Confirms no top-level redesign

55. Engineering Journal

Journal Entry - EA-0012

EA-0012 was created to archive completion of IS-012 - AQELYN Asset & Configuration Governance Engine.

The archive records the expansion of AQELYN into asset and configuration governance. IS-012 defines the structure needed to maintain governed asset inventory, configuration state, baselines, ownership, classification, exposure, criticality, drift findings, and evidence-backed asset governance.

The engineering design preserves the fixed AQELYN repository structure and maintains backward compatibility with previously completed engines.

Lessons Learned

Asset governance must be modeled separately from identity governance and compliance governance. Identities own and administer assets, compliance consumes asset state, and the Asset & Configuration Governance Engine owns the authoritative asset and configuration lifecycle.

Governance Note

EA-0012 follows the master-document publication workflow. The Markdown file is the authoritative source, and PDF/HTML representations are generated from the same content.

56. Examples

56.1 Example Asset

```
asset_id: ASSET-0001
asset_type: cloud_instance
owner: ID-0001
classification: production
lifecycle_state: operational
trust_score: 0.88
criticality: high
```

56.2 Example Configuration Item

```
configuration_id: CFG-1001
asset_id: ASSET-0001
source: cloud_api
baseline: BASELINE-LINUX-HARDENED-v1
observed_state:
ssh_root_login: disabled
disk_encryption: enabled
expected_state:
ssh_root_login: disabled
disk_encryption: enabled
drift_status: no_drift
```

56.3 Example Drift Finding

```
finding_id: DRIFT-2001
asset_id: ASSET-0002
severity: critical
baseline: BASELINE-CLOUD-STORAGE-v2
evidence: evidence://config-snapshot-2026-07-07
reason: public_access_enabled differs from baseline expectation
```

56.4 Example Asset Event

```
{
  "event_type": "configuration.drift.detected",
  "asset_id": "ASSET-0002",
  "configuration_id": "CFG-1002",
  "severity": "critical",
  "baseline_id": "BASELINE-CLOUD-STORAGE-v2",
  "source_engine": "aqelyn_asset_configuration_governance_engine"
}
```

57. Manifest Summary

Archive contents include:

```
README.md
MD/EA-0012.md
PDF/EA-0012.pdf
HTML/index.html
HTML/styles.css
manifest/manifest.json
requirements/Requirements_Matrix.md
traceability/Traceability_Matrix.md
journal/Engineering_Journal.md
diagrams/Architecture.svg
diagrams/Component.svg
diagrams/Workflow.svg
diagrams/EventFlow.svg
diagrams/Integration.svg
examples/example_asset_governance.md
```

58. Final Archive Statement

EA-0012 is the Engineering Archive for IS-012 - AQELYN Asset & Configuration Governance Engine.

It records the completed specification, the architectural decisions, the integration model, the repository impact, the risk posture, verification requirements, acceptance criteria, archive index update, and the engineering publication standard.

```
EA-0012 COMPLETE
IS-012 COMPLETE
NEXT: IS-013
```